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RESEARCH-ARTICLE

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A Study on the Influence Path of AIGC Technology on Creativity Generation of Home Textile Designers

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Abstract

Previous studies have focused on the role of AIGC technology in improving design efficiency and quality, but its specific impact on designers' creativity generation paths has not been explored in depth. This study aims to explore the specific impact of AIGC technology on the creativity path of home textile designers through the SOR theoretical model. Through qualitative and quantitative research methods, this paper analyzes the application of AIGC technology in home textile design, revealing its role in improving design efficiency, optimizing design quality, and promoting human-machine collaboration. The study shows that AIGC technology not only redefines the mode of idea generation, but also provides new impetus for process design innovation. Based on the research results, this paper proposes a strategy to optimize the design process of home textiles, which provides theoretical support and practical guidance for the further application and design practice of AIGC technology, as well as new ideas for future research and industrial applications.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); HCI theory, concepts and models.

Keywords

AIGC, SOR theoretical model, creativity, home textile design

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1 Introduction

With the rapid development of Artificial Intelligence Generated Content (AIGC) technology, the design industry is undergoing a remarkable transformation. AIGC technology enables designers to achieve new breakthroughs in idea generation and process optimization, improve design efficiency, and expand creative thinking through powerful algorithms and rich data resources.[1] In traditional design paths, designers' creativity is often based on past knowledge and experience.[2] With the widespread application of Artificial Intelligence Generated Content (AIGC) technology, the way designers generate ideas is undergoing a major transformation. With powerful algorithms and huge data resources, AIGC technology significantly improves design efficiency and optimizes all phases of idea generation, while giving new impetus to the development of the home textile design industry and promoting innovation and change in design practice. [3] This change in the relationship between technology and designers redefines the traditional path of creativity generation and deserves to be explored in depth. Therefore, this study aims to focus on how AIGC technology affects the creativity generation process of home textile designers, to reveal the interface between technology and creativity in design work, and to propose strategies to optimize the traditional creativity path by using the new technology to promote the continuous innovation and development of the home textile design industry.

2 Current status of AIGC technology research

2.1 AIGC Technology Overview

AIGC, also known as Generative Artificial Intelligence, is an application based on artificial intelligence technology that learns the characteristics and patterns of human language by training large amounts of data and generating a variety of content such as text, images, speech, etc.[4] The generation process of AIGC embodies the in-depth application of AI technology in the field of content creation and brings innovative changes to the field of design by automatically generating high-quality personalized content through the steps of encoding and processing pre-processed data, model

training, decoding and transformation, and quality evaluation. In terms of technical principles, AIGC refers to the use of a GAI algorithmic model to automatically generate task-compliant content through human-provided guidance. The process consists of two core steps: one is to extract target information from human input instructions through semantic parsing and intent recognition; the other is to generate corresponding high-quality content based on the extracted intent information using the GAI model. [5] This technological advancement has laid the foundation for the rapid development of AIGC and driven its wide application in the design field. The training process can be viewed as a minimization-to-maximization problem:

$$\min_G \max_D V(D, G) = E_{x \sim p_{data}(x)} [\log D(x)] + E_{z \sim p_z(z)} [\log (1 - D(G(z)))]$$

The content generation modes of AIGC technology are mainly categorized into text generation, image generation, speech generation, video generation and cross-modal generation. [6] In the field of text generation, OpenAI's ChatGPT model has become a landmark product in the field with its powerful language understanding and generation capabilities. Meanwhile, Google's BERT and T5 models, Facebook's BART and SIMPLIFIED, as well as Baidu's Wenxin Yiyin, Ku Xunfei's Xunfei Starfire, Moon's Dark Side Kimei, and other domestic models, all provide high-quality language services to users. These models use deep learning and natural language processing techniques to generate contextually rich and diverse text content. Currently, AIGC's text generation models are widely used in a variety of scenarios, including but not limited to content creation, automated report generation, customer service, educational support, and entertainment and creative writing. As technology advances, the applications of these models will continue to expand, increasing the efficiency and creativity of language and text processing.

AIGC technology has made significant progress in the fields of image, video and audio, driving innovation in creative content. In image generation, the models represented by DALL-E, Midjourney, and Stable Diffusion, as well as the domestic Tiangong and ERNIE ViL, have been widely used in advertising, design, and other fields thanks to their powerful text-to-image conversion and style migration capabilities. In the video generation field, Synthesia leads the trend of text-to-video conversion, Adobe integrates AIGC technology to enhance video creation tools, while domestic Vidi and "Vision YiSu" demonstrate the innovative strength of long video generation. On the audio side, MuseNet, MusicML, Jukebox and other models realize text-based high-quality audio generation and music creation. In the future, cross-modal generation models such as DALL-E, CLIP, and BLIP will become a trend, further expanding the application boundaries of AIGC technology through multimodal connections and interactions, and realizing seamless integration and innovation in various fields such as vision, speech, and audio.

2.2 Current research status of AIGC technology in the design field

In order to understand the research history and current research status of AIGC technology, Web of Science (WOS) databases were selected as the data sources for this paper, and the data collection

time was September 27, 2024. The foreign literature data were obtained from the WOS core collection, including SCI and SSCI, and the search started at any time. The search strategy was based on a subject search with the search criteria "TS= (AIGC OR Artificial Intelligence) AND TS= (Art design OR Home design OR Home textile design)" and the research direction was "Arts Humanities Other Topics" or 'Art'. After searching, we manually eliminated the literature that was not related to the theme and content by reading and judging, and finally obtained 93 valid literature. In this study, according to the search results, after excluding the misdiagnosed literature, the valid literature was imported into the visual analysis software VOSviewer (1.6.18), and the keyword co-occurrence knowledge map was drawn, see Figure 1.

According to the research hotspots and high-frequency words presented in Figure 1, it can be seen that artificial intelligence has gradually become a research hotspot around innovative design, human-computer intelligence collaboration, user emotional experience, and intelligent generation of content, etc., and its influence is reflected in all aspects of various fields.

In the research literature of WOS, it mainly focuses on combining AIGC technology with design categories, as a way to improve the functionality and convenience for the development of the design industry to provide innovative ideas; [7, 8] in addition, there are also scholars concerned about the AIGC technology on the finished design of the display space to build; [9] the impact of AIGC technology on the designer's creative thinking; [10] AIGC technology to achieve the path of personalized services. [11]

2.3 Characteristics of AIGC in the home textile industry

AIGC technology has shown great potential to enhance design efficiency and creativity in the field of home textile design. Through a survey of 150 home textile designers, the content focuses on three main aspects: (1) whether or not they use AIGC tools when carrying out design work; (2) if they do, which AIGC tools are used most frequently; and (3) whether or not it is important to use AIGC tools when carrying out design work.

This study was conducted in the form of a questionnaire from May 2024, mainly using electronic questionnaires, partially combined with interviews, with a total of 150 questionnaires distributed and the number of valid questionnaires recovered was 150, with a valid recovery rate of 100%. In the questionnaire, the proportion of those aged 18-25 is 51.11% (as shown in Figure 2), 82.3% of home textile designers will use AIGC tools when carrying out home textile design work (as shown in Figure 3), which shows that AIGC tools are widely used in home textile design, and it is considered that the use of AIGC tools when carrying out design work is "very important" and "important". The total percentage of "important" degree is 80.74% (as shown in Figure 4), which shows that home textile designers recognize and attach importance to AIGC tools. For the AIGC tool with the highest frequency of use, 41.48% of home textile designers use Midjourney, 34.81% of home textile designers use ChatGPT, 15.86% of home textile designers use Stable Diffusion, and 7.85% of home textile designers use DALL-E and so on (as shown in Figure 5). It can be seen that these models are

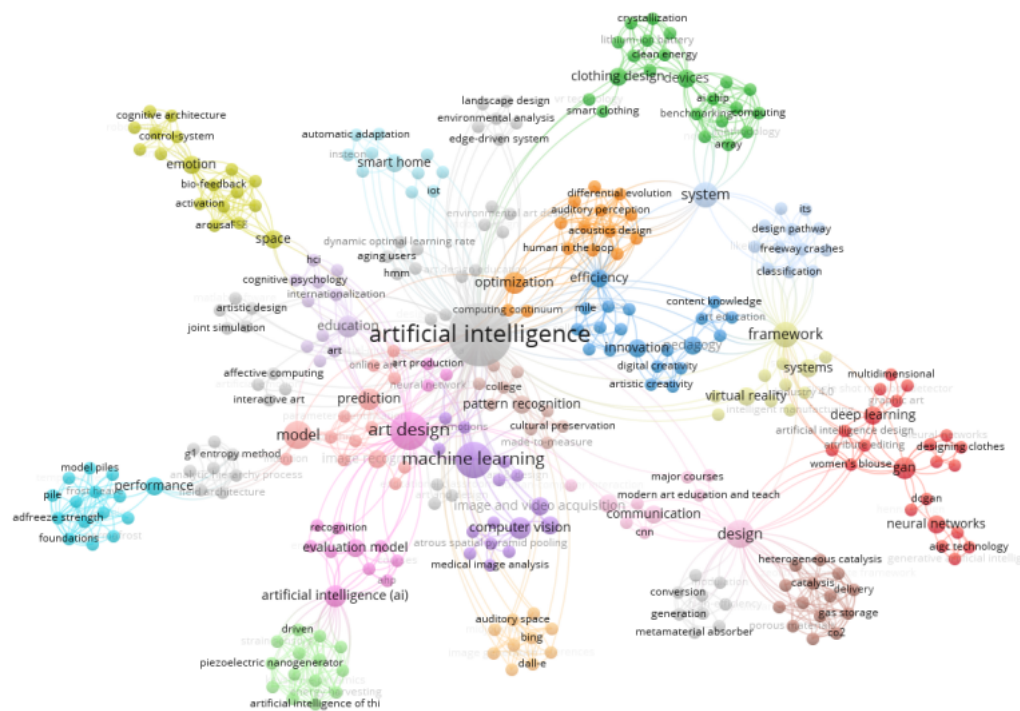


Figure 1: WOS keyword clustering.

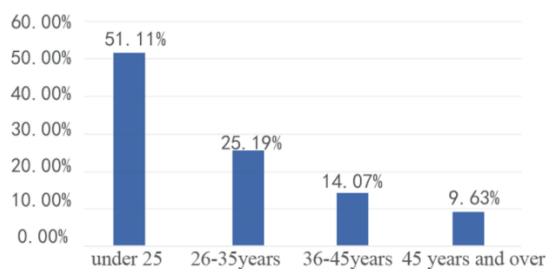


Figure 2: 150 Home Textile Designers Age Distribution Percentage Chart.

gradually becoming the core tools for home textile designers by virtue of their powerful algorithms and innovative capabilities.

The four aigc models most frequently used by home textile designers above demonstrate powerful technical capabilities through their different working principles, and become important tools for home textile designers to carry out their work. chatgpt can realize human intentions from machine learning, neural networks, and Transformer's accumulation of multiple technical models, which are able to obtain "temporal data processing" through "attention mechanisms". midjourney uses deep learning and generative adversarial network technology. "Midjourney utilizes deep learning and generative adversarial network techniques. Midjourney utilizes deep learning and generative adversarial network technology, which continuously improves the quality of the generated images through

mutual confrontation between the generator and the discriminator. The textual descriptions entered by the user are converted by Midjourney into vector form so that they can be understood and processed by the model. Stable Diffusion relies on the combination of a language model and a diffusion model to generate images by gradually adding noise to the image and training the model to remove the noise. This process can be seen as a gradual reduction of a clear image from pure noise.

The application of AIGC in home textile design has remarkable efficiency, interactivity and creativity. Its efficiency is reflected in the fact that AIGC can quickly generate a large amount of text and image materials based on the designer's instructions, which significantly improves the work efficiency of home textile designers. After inputting design requirements, AIGC technology can quickly complete the generation task, helping designers to complete the design scheme in a shorter time, which greatly shortens the design cycle and accelerates the design process. Interactivity is reflected in the interaction between the designer and the AIGC system, where the designer can repeatedly adjust the instructions according to the feedback during the generation process, and the AIGC provides optimized results through instant response. This interactive mechanism not only ensures the flexibility of the design process, but also enhances the accuracy and personalization of the design solution, meeting the diverse needs of designers in home textile design.

In addition, AIGC's creative performance is particularly outstanding. By responding to designers' input instructions and learning styles, AIGC is able to generate creative works of different

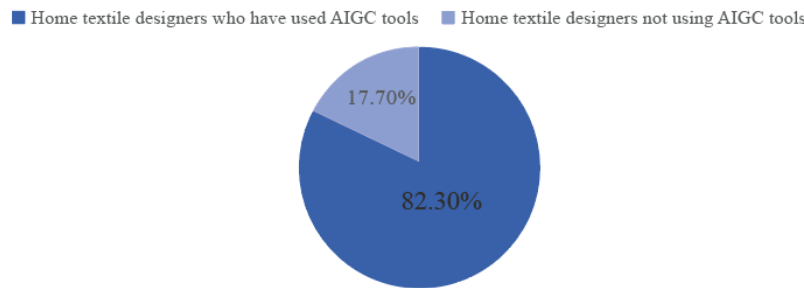


Figure 3: Percentage of home textile designers who have used AIGC tools or not.

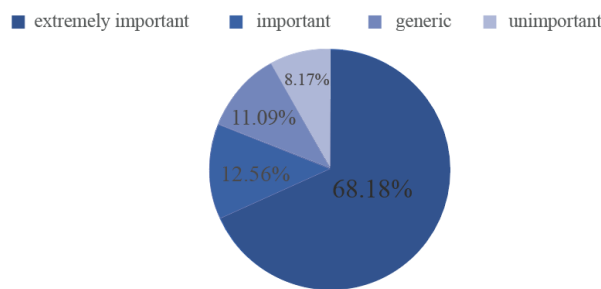


Figure 4: Percentage of home textile designers' usage of AIGC tools.

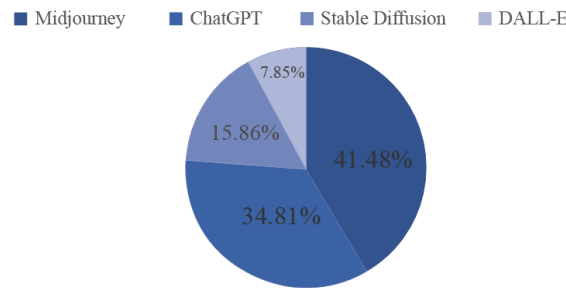


Figure 5: Percentage of AIGC tools most frequently used by home textile designers.

styles, providing designers with a wide range of sources of inspiration. This technology not only helps designers break through the limitations of traditional design thinking, but also supports the innovation of design solutions from multiple perspectives and levels, thus helping home textile designers to cope with the challenges of the complex and changing market and realize the simultaneous enhancement of creative output and commercial value.

3 The Path of Creativity Generation of Home Textile Designers under Traditional Design Models

3.1 S-O-R Theoretical Model and Creativity Generation Theory

S-O-R theory (Stimulus-Organism-Response) was proposed by Mehrabian and Russell (1974) and was first applied in the field of environmental psychology to explain how external environmental stimuli affect an individual's behavioral responses through his or her internal cognitive and affective states (organism).The S-O-R model reveals how situational factors affect an individual's behavioral choices by dividing the environmental stimuli into external

factors and the individual's affective responses, revealing how situational factors influence an individual's behavioral choices. [12, 13]

In recent years, the S-O-R model has been widely used in various fields of research, and new variables have been introduced to form new related models. For example, in the field of consumer behavior, various scholars have combined the S-O-R model with users' consumption intention to explore the influencing factors on users' purchasing desire. Zhang J et al. explain the argument that the environment of scarcity phenomenon increases consumers' fear of missing out on products and thus leads to the influence on consumers' impulsive purchasing. [14] Gao J et al. use the S-O-R model as the theoretical basis to propose that cross-border e-commerce platforms through live broadcasting can indeed increase consumers' overall perceived value and reduce their overall perceived uncertainty, thus enhancing their purchase intention. [15] Peng C, et al. apply the S-O-R model to online shopping behavior, and propose that the environmental stimulus, hedonic shopping value, and utilitarian shopping value can significantly influence online shopping attitudes through an empirical study. [16]

In the field of marketing, Wang Z et al. combined the S-O-R model to conduct an empirical study to explore the positive impact of social media marketing and brand loyalty, and proposed the mediating role of subject happiness. [17] Shi R et al. proposed that short video marketing platforms can stimulate users' heart-flow experience and then generate the willingness to share marketing information. [18] In the field of design, scholars have incorporated the S-O-R model into the research of the design industry. Gao J et al. used the S-O-R model to explore the influence of design perceptual characteristics on the sharing behavior of UI design acceptance, and investigated the mediating role of personal motivation. [19] Ting M P et al. used the S-O-R model in conjunction with technology acceptance to analyze the sharing behavior of a Chinese non-heritage game design, which is a new and unique design. Ting M P et al. combined the S-O-R model with the technology acceptance model to analyze the game design of Chinese non-legacy games by affecting perceived usefulness and ease of use, and then indirectly affecting the churn of game users. [20] Wu W Y et al. used the S-O-R theory to study the impact of store layout design and atmosphere on consumers' shopping intention on websites, and found that environmental stimuli affect customers' shopping behavior through emotional states. [21]

Theories of creativity generation have been heavily researched in two major fields, biology and psychology. Findlay C S et al. argue that creative processes arise as a result of a cognitive level manifestation of making new connections between nodes in an existing knowledge structure and formulating a new series of group selection events. [22] Hunter S T et al. demonstrated that a single knowledge structure such as, for example, schema or associative knowledge is more conducive to the generation of more high-quality ideas. [23] Zhou Dan and Shi Jiannong (2005) suggest that creativity generation is divided into primary and secondary stages: the primary stage involves extensive information gathering and free combination, while the secondary stage filters, optimizes and evaluates initial ideas. In the field of business economics, Asbari M et al. proposed a model to improve the personal innovative behavior of female employees in manufacturing industry by developing psychological capital and personal creativity as mediators.

[24] Slatten T et al. emphasized the important role of leadership practices and work climate on employees' creativity and innovative behavior. [25] In the field of design, Li Y et al. developed a qualitative model of product innovation creativity by integrating cognitive psychology and technology applications into product design from the perspective of creative cognition. [26] Biskjaer M M et al. proposed five approaches to improve creativity and innovation behavior of interaction designers: interaction release, extreme features, metaphorical design, inspiration exploration, and fictional inquiry. [27] Howard Tet al. proposed a four-stage linear model of preparation-illumination-incubation-validation four-stage linear process as a way to identify the stages of creativity generation in the design process. [28] Although the S-O-R model and creativity generation theory have been researched and applied in many fields, there is a lack of research on creativity generation of designers in the field of home textile design. Exploring the different paths of creativity generation of home textile designers under the traditional model and under the intervention of AIGC can more fully reflect the actual value of AIGC technology for the home textile design industry.

The combination of S-O-R theory and creativity generation theory can effectively explain how designers generate innovative designs in complex external environments. The reason for choosing S-O-R theory and creativity generation theory in this study is that the creative process of home textile designers is highly dependent on the influence of external market changes and technological innovations, etc. The S-O-R model can systematically explain how the external environment influences the designers' creative thinking and design decisions through their cognitive and affective states. Meanwhile, the creativity generation theory reveals how designers gradually generate innovative designs during the information processing process, from inspiration collection to solution optimization. Therefore, combining these two theories helps to understand more comprehensively the path of creativity generation of home textile designers in the traditional design mode, and provides theoretical basis for innovative practice in the field of home textile design.

As shown in the Figure 6, the creativity generation path based on the S-O-R model demonstrates the dynamic relationship among stimulus, organism and response and their complex interactions in the process of home textile designers' creativity generation.

3.2 Path of Creativity Generation for Home Textile Designers under Traditional Mode

The creativity generation process of home textile designers in the traditional design mode can be explained by the S-O-R theory (Stimulus-Organism-Response) and the three-stage model of creativity generation. The creativity generation process of designers is divided into three stages: external stimulation, internal cognitive processing and final response. In the external stimulus stage (S), designers obtain design inspiration through market research, trade shows, industry reports, and direct feedback from customers. Factors such as market trends, user needs and technological changes in the external environment serve as stimuli to stimulate designers' perceptual and cognitive responses. Through these information resources, designers are able to grasp the dynamics of the industry and predict the future needs of consumers. [29] This stage is not

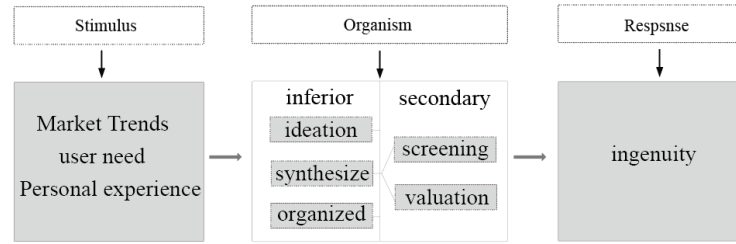


Figure 6: Creativity generation map for home textile designers under S-O-R theory.

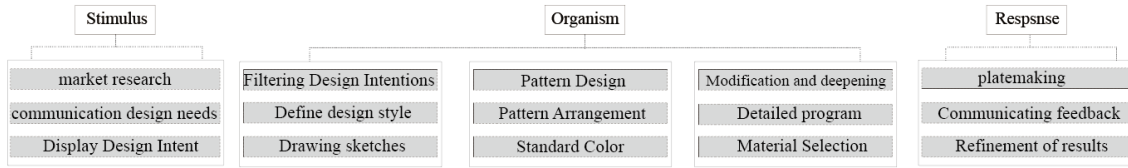


Figure 7: Path of Creativity Generation for Home Textile Designers under Traditional Mode.

only a passive reception of information, but the designer also needs to actively analyze and select information about design-related trends in order to form a preliminary design concept.

Entering the internal cognitive processing stage, the designer optimizes and justifies the preliminary design solution. This stage corresponds to the second stage of the theory of creativity generation, the integration and screening of information. [30] The designer analyzes the feasibility of the design solution in detail by evaluating the materials, processes and production costs.[31, 32] At the same time, designers need to combine the information obtained from customer and market feedback to make gradual modifications and adjustments to the initial ideas to ensure that the design solutions not only have market value but can also be technically realized.[33]

In the final reaction stage (R), the designer's creativity is fed back and evaluated through concrete design realization. This stage corresponds to the third stage of creativity generation, i.e. optimization and realization of the solution. [34] Designers further refine the design details through sketching, material testing, and sample production to ensure that the design solution is balanced in terms of aesthetics and functionality. At the same time, the designer needs to work closely with the production team to ensure that the design meets the technical standards for production and is successfully put into mass production. This stage is not only the verification of the design idea, but also the final assessment of the whole idea generation process.

Figure 7 shows the path of creativity generation of home textile designers under the traditional model demonstrates the complex interaction between the external environment and the designers' internal cognitive processing. Although the idea generation in this mode relies on the market and customer demand orientation, the individual creativity of designers still plays a key role in the process

of information integration, inspiration generation and solution optimization.

4 The Intervention of AIGC Technology in the Path of Creativity Generation of Home Textile Designers

4.1 AIGC Intervention in External Stimulus Conditions

AIGC (Artificial Intelligence Generated Content) technology has dramatically changed the way designers obtain external stimuli in the process of generating creativity for home textile designers. Under the traditional design mode, designers mainly rely on periodic market reports, process and consumer research to obtain information on market trends and user needs. These methods are not only short-lived, but also suffer from the problems of lack of timeliness and information lag, and the introduction of AIGC technology provides designers with a more dynamic and real-time way to obtain market information, which effectively reduces the problem of information lag and improves the efficiency of information processing. Through AIGC technology, designers are able to filter out key market trends and consumer behavior patterns from a wide range of real-time data. This information processing capability not only improves the breadth and depth of information acquisition, but also ensures the timeliness and relevance of the data. In addition, AIGC enhances its trend prediction capability through the analysis of historical data. AIGC technology can predict future design trends based on historical market trends and consumer behavioral patterns, which can help designers to make smarter designs. This prediction capability provides more accurate design directions for designers, and helps designers to make smarter designs by providing more accurate design directions. AIGC technology can predict future

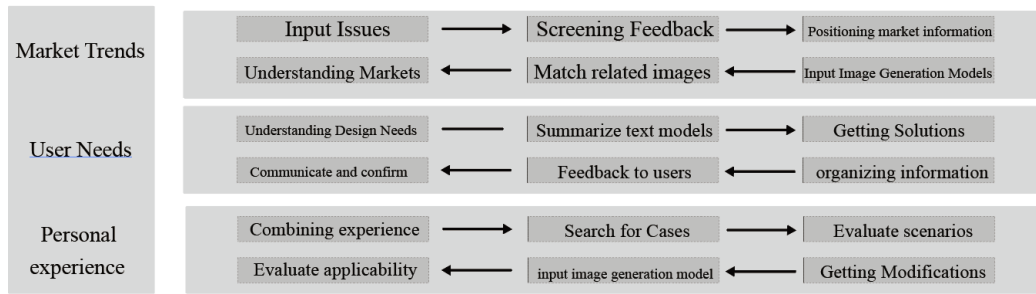


Figure 8: Map of AIGC interventions in stimulation.

design trends based on historical market trends and consumer behavior patterns, helping designers make smarter designs.

AIGC technology also has significant value in the analysis of user needs and customized design. Through fast natural language processing technology, AIGC is able to summarize and summarize users' personalized needs, and further generate design solutions that meet users' preferences. [35] The application enables designers to capture customer needs more accurately and respond quickly to those needs, improving the efficiency and accuracy of the design process. In terms of the designer's personal experience, AIGC is able to help designers optimize their design decisions by analyzing design history data. Designers are not only able to conduct a comprehensive analysis by combining past experience and current project requirements, but also use AIGC's recommendations to generate more innovative design solutions. This customized experience analysis and suggestion generation enhances the designer's decision-making ability in the innovation process and improves the design's market sense and technology sense.

AIGC technology accomplishes the creativity generation path of home textile designers through the customized processing of market information, user needs and design experience (as shown in Figure 8). Its intervention in external stimulus acquisition, prediction ability enhancement, design feedback optimization and personalized customization significantly enhances designers' creativity generation efficiency and market competitiveness.

4.2 AIGC Intervention in Organismal Transformation

In the traditional design paradigm, as the primary stages of transformation include conceptualization, synthesis and organization. This stage is the key stage of idea generation, as external stimuli are processed through this final stage, which directly affects the performance of creativity. In the conceptualization stage, the designer's home task transforms abstract user needs into concrete design concepts. By interacting with AIGC technology, designers can quickly obtain detailed information about design styles, color usage, light and shadow effects, and other design elements, thus providing a sophisticated theoretical basis for styling. [36] and the responsiveness and customization of AIGC technology greatly improves the designers' ability to comprehend and deal with complex design styles.

In the synthesis stage, designers utilize AIGC to further refine the design scheme based on the information obtained in the preliminary stage. With the help of AIGC technology, designers can quickly access cases related to the current design and transform them into visualized design solutions. In this process, AIGC's image generation technology provides designers with a tool to efficiently generate multiple visual sketches in a short period of time and repeatedly adjust the design parameters to ensure the diversity and innovation of the design solutions. [37] The synthesis process reduces the repetitive work in traditional manual design and improves design efficiency and creative quality.

In the organization stage, designers filter and optimize the pre-conceived ideas and synthesized materials to finally form the most suitable design solutions. The AIGC image generation model provides a basis for comparison and selection, allowing designers to quickly generate diverse design combinations by inputting different design elements and carefully evaluating their visual effects. This stage not only reduces the manual workload associated with traditional design, but also provides designers with more creative space, which further enhances the personalization and innovation of the design. [38]

In the screening and evaluation stage, designers can use AIGC technology to optimize and present the design solution. The text and image models generated by AIGC can make the design solution appear more professional and guided, which improves the efficiency of communication with customers. After providing user feedback, designers can quickly use AIGC to make local adjustments and optimizations based on the feedback to further meet the user's individual needs, thus accelerating the design decision process.

Ultimately, after the design solution is selected by the user, the designer also needs to conduct a comprehensive assessment of the solution's suction power, including production costs, technical suction power and market prospects. Through AIGC's analysis and simulation functions, the designer was able to get a comprehensive understanding of the potential and challenges of the solution in actual production, and made data-supported decisions (as shown in Figure 9). After several rounds of optimization and adjustment, the final designer determined the design scheme and completed the whole process of idea generation.

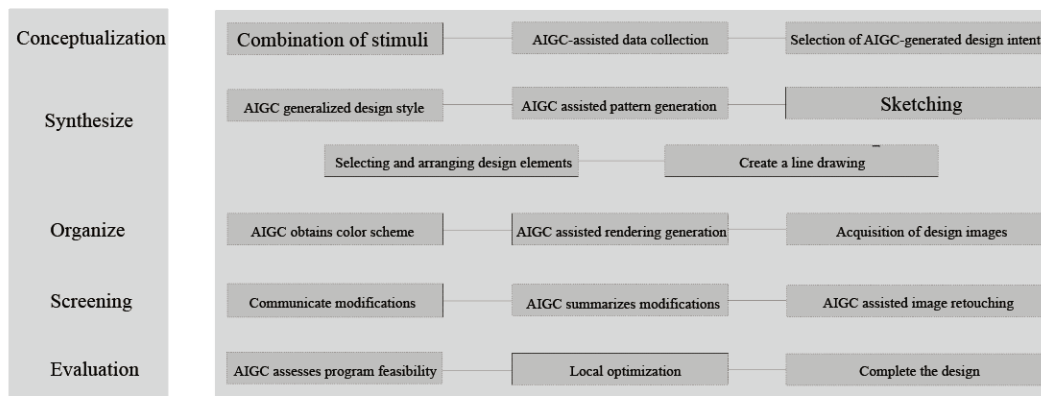


Figure 9: Map of AIGC interventions in organism.

4.3 Home textile design using Stable Diffusion as an example

Stable diffusion as an open source model, although its operation difficulty requires users to have a certain degree of experience, but the generated image effect is realistic, high quality and can be customized model of the function enriches the user's personalized and innovative creation needs. Users can generate a new image based on the uploaded image and customize the image effect by modifying the prompt, and they can also use text to generate the image by inputting the relevant positive and negative prompts to guide the model to generate the relevant image according to the specific needs. The model adopts majiacMIXrealistic, the algorithm uses ampler:Euler a, after practicing stable diffusion home textile design field can realize home textile product line draft design, home textile product floor plan design, home textile product effect design.

4.3.1 Home textile product line drawing design. Line drawing in home textile design has the significance of the important link, is absorbed in the stimulus to the formation of the design concept after the more detailed design concept of the initial presentation, which affects the advancement of the later design. Home textile designers will be their own design needs through the text way to input or to the preliminary sketch as a sample for drawing, through the parameter settings and adjustments stable diffusion can quickly respond to the design of the line drawing, designers can also be based on this to modify the adjustments again, improve the efficiency of the work of drawing the line drawing at the same time to obtain a more accurate detail of the image effect(as shown in Figure 10). The designer can also make changes and adjustments based on this, improving the efficiency of drawing line drawings while obtaining more precise details of the image effect.

4.4 Home textile product floor plan design

According to the pre-determined line drawing program, you can enter some keywords about the category, flower shape, color, etc., you can get a complete sense of detail and a sense of aesthetics of the floor plan, the floor plan compared to the line drawing has a more complete design content, you can see the effect of the finished

product from the effect of the display(as shown in Figure 11). For the design problems that existed in the pre-draft design stage in this stage through the deepening of the design to have a more intuitive display of the effect of adjustment and modification. In this stage, stable diffusion may not be able to accurately identify some specific professional vocabulary in home textile design, designers can upload pictures to give stable diffusion templates to make the picture effect in line with the designer's psychological expectations.

4.5 Home textile products effect design

In the traditional home textile design process, after determining the plane effect diagram, home textile designers need to understand the effect of its use in reality, usually will use the prototype pasting way. This process requires time-consuming and labor-intensive repetitive operations, and lacks diversity in the matching of scenes. In stable diffusion can be directly converted into intuitive effect diagrams, applied to home textile products, presenting a more three-dimensional visual effect, but also through the adjustment of parameters to add the corresponding prompt to make the overall display of home textile design effect is more holistic and a sense of atmosphere, in order to enhance the beauty and attractiveness of home textile products (as shown in Figure 12).

4.6 Future development models for human-computer collaboration

After passing through the primary stage of external stimulation and organism transformation, home textile designers can obtain external stimulation faster and more accurately with the help of AIGC, and break through the inherent thinking at the cognitive level to effectively stimulate creativity at the final stage. The traditional way of generating ideas for home textile design largely relies on the designer's personal thinking and experience, and the effectiveness of external stimulation and organism transformation is limited. However, the introduction of AIGC technology provides designers with more design inspiration and innovative ideas by learning the commonality of design factors and focusing on the formal characteristics of design. [39] This technology not only simplifies

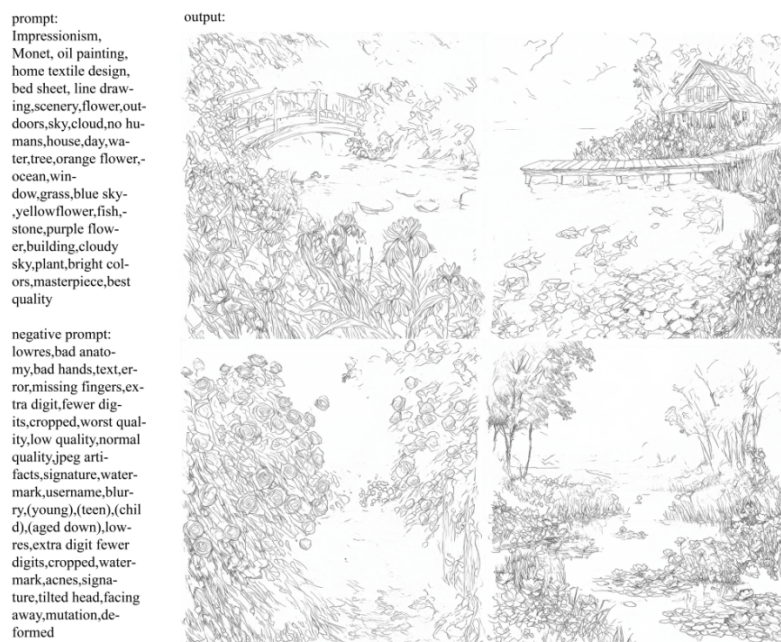


Figure 10: Stable diffusion Home textile products line drawing design.

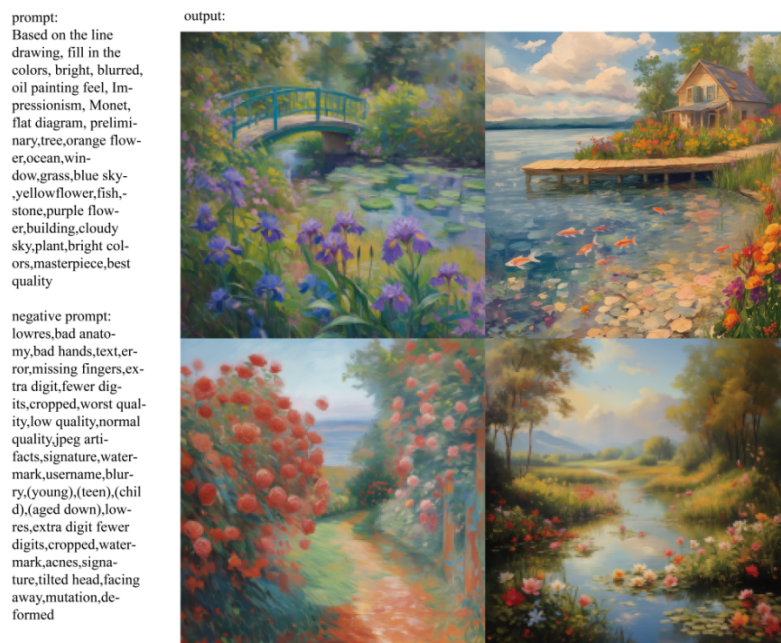


Figure 11: Stable diffusion Home textile products floor plan design.

the creative process, but also enables home textile designers to raise the bar of creativity in an increasingly competitive industry.

Although AIGC technology is capable of generating images quickly and with high quality, this automated feature has raised concerns and a sense of crisis among designers about replacing

design work. In recent years, the question of whether the future role of designers will be replaced by AIGC has become a hot topic. However, from the perspective of human-machine manipulators, AIGC should be seen as a creative tool for home textile designers rather than a replacement. In the human-machine collaborative

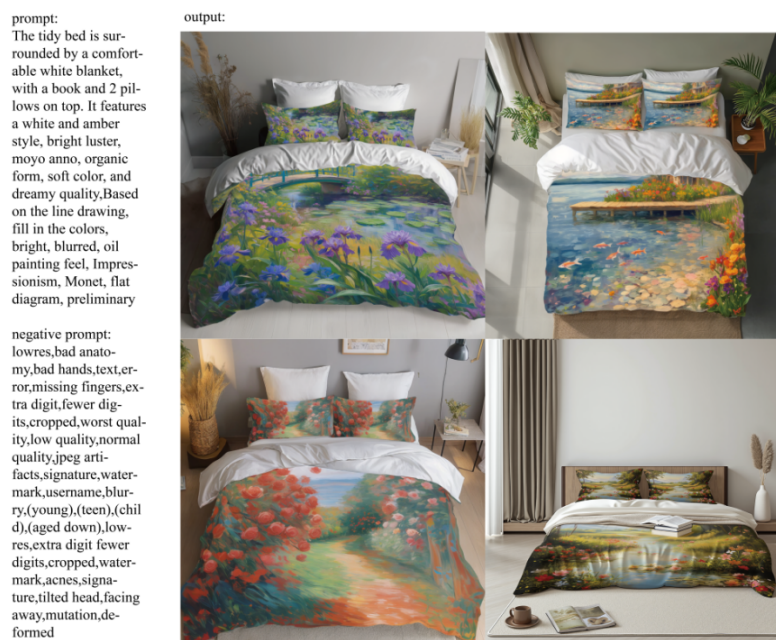


Figure 12: Stable diffusion Home textile products effect design.

work mode, the AIGC's response depends entirely on the designer's instructions and guidance. Without the designer's participation, the AIGC cannot perform creative activities on its own, and even the generated content must be adjusted and modified by the designer to ensure that it meets the design requirements. Through interaction and training with the AIGC model, the designer can gain more inspiration in the process of thought dispersion and finally realize the output of creativity.

By continuously interacting with designers, the AIGC model also gradually understands designers' preferences and styles to better support their design work. This human-computer interaction mode is gradually becoming an important way for designers to break through the limitations of their own thinking. In the future, with the continuous advancement of AIGC technology, more intelligent models will appear. Designers need to rethink the core creativity of the home textile industry, and through cooperation with AIGC, continue to explore and practice to realize the best integration of technology and design thinking. [40] This integration will help designers export their creativity more effectively and become more competitive in the industry.

5 Summary

This paper explores the role of AIGC intervention in the stages of external stimulation, organism transformation, and final creativity generation, as well as the development model of human-computer collaboration in future design. In the traditional design model, designers' creativity relies on personal experience and limited processing of external information, while AIGC provides designers with more efficient tools and support through real-time data processing, intelligent feedback, and autonomous learning capabilities.

AIGC not only improves designers' speed in detecting market trends and user needs, but also significantly shortens the design iteration cycle and trial-and-error period through design generation models and intelligent feedback mechanisms. AIGC not only improves designers' access to market trends and user needs, but also significantly shortens design iteration cycles and trial-and-error costs through design generation models and intelligent feedback mechanisms. In the human-computer collaboration mode, AIGC acts as an assistive tool for designers to enhance their creative expression and help them gain more inspiration and innovation opportunities in complex design tasks.

With the further development of AIGC technology, human-computer collaboration modeling will play a more important role in home textile design. Designers must deeply explore the integration of future technology and creativity, and find more efficient design methods and ways through continuous practice and innovation. Finally, AIGC is not only a tool for designers, but also an important engine for promoting creativity in the home textile industry, and injecting new vitality into the sustainable development of the industry.

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